FEA notation, assembled in order

residual, test, trial, shape function, Galerkin, the dot product, and Galerkin orthogonality

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Two running examples, so nothing stays abstract:

Example A	$-u'' + q(x)u = f$ on $(-\pi, \pi)$, periodic, with $q = 3 - \sin x - \sin 2x - \cos x$, $f = 2e^{\sin x}e^{\cos x}$, exact $u = e^{\sin x}e^{\cos x}$. Used for the numbers.
Example B	$-u'' = 2$ on $(0,1)$, $u(0) = u(1) = 0$, exact $u = x(1-x)$. Two hat functions, one unknown. Worked by hand in §11.
Example C	The same BVP, but the space is $V_h = \text{span}\{\sin \pi x\}$ — still one unknown, but u is *not* in the space, so the error is visible. The whole chain is walked on it in §12.

1 The residual R — the thing you are trying to kill

Move everything to one side and feed it *any* candidate function w:

$$\boxed{R(w)(x) = -w''(x) + q(x)\,w(x) - f(x)}$$

- R measures how badly w fails the equation, **point by point**.
- $R(u) \equiv 0$. That is exactly what "u is the exact solution" means.
- For anything you can actually compute, $R \neq 0$ somewhere.

Why w, not u

- u is reserved: it means *the* exact solution, one specific function.
- R must accept *any* candidate. $R(u)$ only ever describes the answer you do not have — and equals zero, so it says nothing.
- w is a placeholder: "whatever function you hand me".

Reading the brackets

R is an **operator**: a function goes in, a function comes out.

written	what it is	compare
R	the operator. "the residual"	D, differentiation
$R(w)$	a **function** of x	$D(w) = w'$
$R(w)(x)$	a **number**: its value at the point x	$D(w)(x) = w'(x)$

- $R(x)(w)$ is **wrong**. The order is not free: you must supply the function first — there is nothing to differentiate until you do — and only then the point.
- $R(x)$ alone is loose shorthand. It hides *which* candidate produced it. Fine in prose when the candidate is obvious, ambiguous in a definition.

Same operator, three different candidates (Example A):

x	$R(u)(x)$ *exact*	$R(w_1)(x)$ $w_1 = 1$	$R(w_2)(x)$ $w_2 = \cos x$
-2.0	0	3.037	-2.433
-1.0	$4 \cdot 10^{-16}$	2.731	1.335
0.0	0	-3.437	-2.437
1.0	0	-7.255	-7.041
2.0	0	-0.011	-5.049

- $R(u)$ is the zero *function*; $R(w_1)$ is a different function, not zero; $R(w_1)(0) = -3.437$ is a single *number*.
- $R(w_1)(2) = -0.011$: a wrong candidate can be accidentally right at a point. R small *somewhere* means nothing.

2 Why you cannot kill R everywhere

- "$R(w)(x) = 0$ at every x" is **infinitely many** conditions, one per point.
- You have **finitely many** unknowns.
- So do not kill R everywhere. Kill a *finite number of weighted averages* of it.

3 Test function — the probe

Pick a weight function v and demand

$$\int R(W)\, v \, \mathrm{d}x = 0.$$

- Meaning: *R has no component along v*. Not that R is zero — only that v cannot detect it.
- v is the **test function**: a probe, a measuring device.
- **One test function = one equation.** Choose $M + 1$ of them and you get $M + 1$ equations for $M + 1$ unknowns. That is the entire design.

4 Trial function — the candidate

The thing you solve *for*, restricted to a chosen form:

$$U(x) = \xi_0 \varphi_0(x) + \xi_1 \varphi_1(x) + \cdots + \xi_M \varphi_M(x).$$

trial	what you build the answer out of
test	what you probe the leftover with

Watch out: they are not opposites

Trial and test are two *jobs*, not two ends of something. They are not perpendicular to each other — and in the method below they are drawn from the same set of functions.

5 "Galerkin" — one specific choice

You must decide which functions to use as tests:

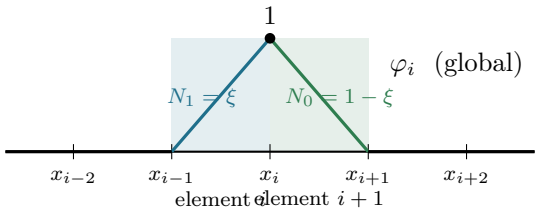
Galerkin	test space = trial space
Petrov–Galerkin	test space ≠ trial space

- **That is the entire content of the word "Galerkin".** Nothing more.
- Both examples here are Galerkin: tests and trials both come from $\text{span}\{\varphi_0, \ldots, \varphi_M\}$.

6 N versus φ versus "shape function"

Same objects, two levels of bookkeeping.

- N — **local, per element.** On one element mapped to $0 \le \xi \le 1$, a 2-node linear element has $N_0(\xi) = 1 - \xi$, $N_1(\xi) = \xi$. Two functions on one element.
- φ_i — **global, per node.** Hold every node at zero, lift node i to 1, join with straight lines.
- **Relation:** φ_i is just the local N's that touch node i, stitched together.
- **"Shape function"** is the engineering word — usually N (local), sometimes φ (global). Same functions.



On the element *left* of node i, node i is the right node, so it carries $N_1 = \xi$. On the element *right* of it, node i is the left node, so it carries $N_0 = 1 - \xi$. Zero on every other element.

Two properties are the reason this basis is used:

- $\varphi_i(x_j) = 1$ if $i = j$, else 0. So in $U = \sum_j \xi_j \varphi_j$ each coefficient ξ_j **is** the nodal value $U(x_j)$ — the unknowns are the displacements themselves, no transformation.
- φ_i is nonzero on only **two** elements, so φ_i and φ_j overlap only for neighbours. That is where sparsity comes from.

7 The dot product — and where it comes from

- **Vectors:** $\boldsymbol{a} \cdot \boldsymbol{b} = a_1 b_1 + a_2 b_2 + \cdots$ — multiply matching components, add.
- **Functions:** $\int v\, w \, \mathrm{d}x$ — multiply at each x, add over x. Same operation, continuous index.

The pairing you actually use is **not invented**. It falls out of the test equation after one integration by parts:

$$\int R(U)\, v \,\mathrm{d}x = 0$$

$$\int \big(-U'' + qU - f\big) v \,\mathrm{d}x = 0$$

$$\underbrace{\int \big(U'v' + q\,U\,v\big)\,\mathrm{d}x}_{\text{call this } \langle U,\, v\rangle} \;-\; \underbrace{\big[U'v\big]}_{=0 \text{ at the ends}} \;-\; \int f\, v \,\mathrm{d}x = 0$$

$$\boxed{\langle v,\, w\rangle := \int \big(v'w' + q\,v\,w\big)\,\mathrm{d}x} \qquad \text{(Example B has } q = 0\text{, so } \langle v,\, w\rangle = \int v'w'\,\mathrm{d}x\text{)}$$

- **It is a genuine dot product.** The integrand is $\big[v', \sqrt{q}\,v\big] \cdot \big[w', \sqrt{q}\,w\big]$ — an ordinary 2-component dot product at each x, summed over x.
- Symmetric; linear in each slot; and $\langle v,\, v\rangle = \int \big[(v')^2 + qv^2\big]\,\mathrm{d}x > 0$ unless $v \equiv 0$. Exactly the properties of $\boldsymbol{a} \cdot \boldsymbol{b}$.

Intuition: what integrating by parts bought

The original R needs U''. After moving one derivative onto v, only U' appears. That is what makes a piecewise-linear U — which has no second derivative at all — a legal candidate.

8 Orthogonality — the generic word

- $\langle a,\, b\rangle = 0$ means a **has no component along** b. Perpendicular.
- Exactly as with vectors. Nothing FEM-specific about the word itself.

9 Testing with each φ_i *is* $\boldsymbol{K\xi = b}$

Use $v = \varphi_i$, one at a time, $i = 0, \ldots, M$:

$$\langle U,\, \varphi_i\rangle = \int f\,\varphi_i\,\mathrm{d}x \qquad \text{one equation per test function}$$

$$\Big\langle \textstyle\sum_j \xi_j\varphi_j,\ \varphi_i \Big\rangle = \int f\,\varphi_i\,\mathrm{d}x \qquad \text{substitute } U = \textstyle\sum_j \xi_j\varphi_j$$

$$\sum_j \xi_j\,\langle \varphi_j,\, \varphi_i\rangle = \int f\,\varphi_i\,\mathrm{d}x \qquad \text{linear in the first slot}$$

which is row i of a matrix system:

$$\boxed{K_{ij} = \langle \varphi_j,\, \varphi_i\rangle, \qquad b_i = \int f\,\varphi_i\,\mathrm{d}x, \qquad \boldsymbol{K\xi = b}}$$

- $\boldsymbol{K}$ is the **table of pairings of the basis functions with each other**. Checked on Example A: $\max_{ij}\big|K_{ij} - \langle \varphi_j,\, \varphi_i\rangle\big| = 2.2 \times 10^{-16}$.
- Its diagonal is $\langle \varphi_i,\, \varphi_i\rangle$ — large and positive ($4.879,\ 4.722,\ 4.513, \ldots$ on a 7×7 case).
- So the φ's are emphatically **not** orthogonal to one another. If they were, $\boldsymbol{K}$ would be the zero matrix and there would be nothing to solve.

10 Galerkin orthogonality

- The exact u satisfies the identity for *every* admissible v, including each φ_i: $\langle u, \varphi_i \rangle = \int f\varphi_i \, \mathrm{d}x$.
- Row i of your system says $\langle U, \varphi_i \rangle = \int f\varphi_i \, \mathrm{d}x$.
- Subtract. **The load cancels.** With the error $e := u - U$:

$$\boxed{\langle e, \varphi_i \rangle = 0, \qquad i = 0, 1, \ldots, M}$$

Galerkin orthogonality	it holds because the tests were taken from the trial space — which is how U was built the pairing is zero

Why the two sides are equal — they get there differently

The subtraction looks empty: two identical numbers giving zero. The content is in *why* each side reaches that number.

$\langle u, \varphi_i \rangle = \int f\varphi_i$	because u solves the problem against *every* test function, and φ_i happens to be one of them. u had no choice; nobody arranged this.
$\langle U, \varphi_i \rangle = \int f\varphi_i$	because you **forced** it. This is the equation you imposed: $\langle U, \varphi_i \rangle = \boldsymbol{K}\boldsymbol{\xi} = \boldsymbol{b} = \int f\varphi_i$.

- One side is a *property* of the true solution; the other is a *constraint you built in*. They reach the same number from opposite directions.
- You did not know $\langle u, \varphi_i \rangle$ at the start. You could compute $\int f\varphi_i$ from f alone — so you learned one true number about u without knowing u. Then you chose $\boldsymbol{\xi}$ to reproduce it.
- **That is the accomplishment:** $M + 1$ quantities of the unknown solution, matched *exactly*, paid for with f alone.
- Why rewrite it as $\langle e, \varphi_i \rangle = 0$? Same fact, usable form — compare $x = 5$ with $x - 5 = 0$. The second is what you substitute into other equations, and it is exactly what deletes a term later:

$$\|e\|_E^2 = \langle e, u - U \rangle = \langle e, u - v \rangle + \underbrace{\langle e, v - U \rangle}_{=0,\ v-U \in V_h} = \langle e, u - v \rangle.$$

Intuition: one scale

You own one scale; it reports one number.

- From the spec sheet (f) you know the true object must read 1.273 — without ever putting it on the scale.
- You tune your model until *it* reads 1.273.
- Now weigh the difference between them: **0.000**.

The difference is not absent — the two objects genuinely differ. It is *unmeasurable by that scale*. Your instrument cannot tell them apart.

Watch out: what is orthogonal to what

Galerkin orthogonality is **the error against the test functions**, $\langle e, \varphi_i \rangle = 0$. It is *not* trial against test, $\langle \varphi_i, \varphi_j \rangle$ — those are the entries of $\boldsymbol{K}$, and they are nonzero. Two different

objects: $\boldsymbol{K}$ is built from the nonzero one, the theorem is about the zero one.

Measured on Example A, $M = 24$, so 25 basis functions:

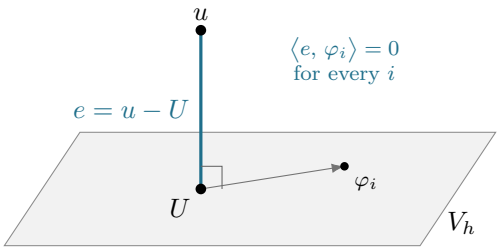
$\max_i \lvert\langle e,\, \varphi_i\rangle\rvert$ over all 25	7.2×10^{-14}	(machine zero)
$\langle e,\, \sin 2x\rangle$	0.156	
$\langle e,\, \cos 3x\rangle$	-0.125	
$\langle e,\, e\rangle$	0.171	

- **What it does not say.** $\langle e,\, \varphi_i\rangle = 0$ is $M+1$ numbers; e is a whole function. $M+1$ numbers cannot determine it — which is why e stays nonzero everywhere while every pairing is exactly zero. You learn e's *direction*, never its *size*.

Intuition: the error is not small — it is invisible

Think of the $M+1$ basis functions as $M+1$ **sensors**. The only measurement the solver ever performs on a function is $\langle \cdot,\, \varphi_i\rangle$ — that is what assembling a row does. All 25 sensors read zero on e; an outside probe reads 0.156. The error sticks straight out of the space the mesh can represent.

It has to: if any sensor had read nonzero, one row of $\boldsymbol{K\xi} = \boldsymbol{b}$ would be violated, and $\boldsymbol{\xi}$ would not be the solution.



V_h is everything the mesh can represent; the φ_i are its axes. e has zero component along every axis.

11 Everything above, by hand

Check it by hand: Example B, one unknown

$-u'' = 2$ on $(0,1)$, $u(0) = u(1) = 0$. Exact solution $u = x(1-x)$. Here $q = 0$, so $\langle v,\, w\rangle = \int_0^1 v'w'\,\mathrm{d}x$.

Mesh: two elements, $h = \frac{1}{2}$, one free node at $x_1 = \frac{1}{2}$. So $V_h = \operatorname{span}\{\varphi_1\}$ — exactly *one* unknown ξ_1 and *one* test function.

$$\varphi_1(x) = \begin{cases} 2x, & 0 \le x \le \frac{1}{2} \\ 2(1-x), & \frac{1}{2} \le x \le 1 \end{cases} \qquad \varphi_1'(x) = \begin{cases} +2, & 0 < x < \frac{1}{2} \\ -2, & \frac{1}{2} < x < 1 \end{cases}$$

The matrix (one entry).

$$K_{11} = \langle \varphi_1,\, \varphi_1\rangle = \int_0^1 (\varphi_1')^2\,\mathrm{d}x = 2^2 \cdot \tfrac{1}{2} + (-2)^2 \cdot \tfrac{1}{2} = \boxed{4}$$

The load (one entry). $\int_0^1 2\varphi_1\,\mathrm{d}x = 2 \times$ (area of the hat) $= 2 \times \frac{1}{2} \cdot 1 \cdot 1 = \boxed{1}$

Solve. $4\xi_1 = 1 \Rightarrow \xi_1 = \frac{1}{4}$. And indeed $u(\frac{1}{2}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ — exact at the node.

Now check Galerkin orthogonality by hand. With $u' = 1 - 2x$ and $U' = \pm\frac{1}{2}$:

$$\langle u,\, \varphi_1 \rangle = \int_0^{1/2} (1-2x)(2)\,\mathrm{d}x + \int_{1/2}^{1} (1-2x)(-2)\,\mathrm{d}x = \tfrac{1}{2} + \tfrac{1}{2} = 1$$

$$\langle U,\, \varphi_1 \rangle = \int_0^{1/2} (\tfrac{1}{2})(2)\,\mathrm{d}x + \int_{1/2}^{1} (-\tfrac{1}{2})(-2)\,\mathrm{d}x = \tfrac{1}{2} + \tfrac{1}{2} = 1$$

$$\Longrightarrow \quad \langle e,\, \varphi_1 \rangle = 1 - 1 = \boxed{0}$$

But the error is not zero. $e = u - U$:

x	$u(x)$	$U(x)$	$e(x)$
0.25	0.1875	0.1250	0.0625
0.50	0.2500	0.2500	0
0.75	0.1875	0.1250	0.0625

$$\langle e,\, e \rangle = \int_0^1 (e')^2\,\mathrm{d}x = \tfrac{1}{12} \approx 0.0833 \;\neq\; 0, \qquad \langle e,\, \sin \pi x \rangle = 0.273 \;\neq\; 0.$$

Read it: $\langle e,\, \varphi_1 \rangle = 0$ exactly, while $\langle e,\, e \rangle = \frac{1}{12}$ and $\langle e,\, \sin \pi x \rangle = 0.273$. The error is a real, sizeable function — it is only the *one* probe the mesh owns that cannot see it.

12 The whole chain, walked on one unknown

Check it by hand: Example C — one unknown, and a visible error

Same BVP as Example B: $-u'' = 2$ on $(0, 1)$, $u(0) = u(1) = 0$, exact $u = x(1 - x)$. But now take the space

$$\varphi(x) = \sin \pi x, \qquad V_h = \operatorname{span}\{\varphi\}, \qquad U = \xi\, \varphi.$$

One unknown ξ. Note $u = x(1-x)$ is *not* a multiple of $\sin \pi x$, so there will be a genuine error — which is the point. Here $q = 0$, so $\langle v,\, w \rangle = \int_0^1 v'w'\,\mathrm{d}x$.

→ **R is the leftover.** $R(U) = -U'' - 2 = \xi\pi^2 \sin \pi x - 2$, an honest function of x:

ξ	$R(0.25)$	$R(0.50)$	$R(0.75)$
0.0000	−2.0000	−2.0000	−2.0000
0.2580	−0.1995	0.5464	−0.1995
1.0000	4.9789	7.8696	4.9789

→ **Cannot kill it everywhere.** $\xi\pi^2 \sin \pi x - 2 = 0$ needs $\xi = 2/(\pi^2 \sin \pi x)$ — a *different* ξ at every x. One unknown, infinitely many demands.
→ **Kill $M + 1 = 1$ weighted average.** One unknown, so buy exactly one equation.
→ **The weight is the test function; Galerkin takes it from the trial space.** The trial

space is span$\{\sin \pi x\}$, so test with $v = \sin \pi x$:

$$\int_0^1 R(U) \sin \pi x \, \mathrm{d}x = 0 \quad \Longrightarrow \quad \xi \underbrace{\int_0^1 \pi^2 \sin^2 \pi x \, \mathrm{d}x}_{4.934802} = \underbrace{\int_0^1 2 \sin \pi x \, \mathrm{d}x}_{1.273240} \quad \Longrightarrow \quad \xi = 0.258012.$$

Look back at the table: $\xi = 0.2580$ is the row where R is nowhere zero, but has zero weighted average.
→ **Integrate by parts; the dot product appears.** $\int (-U'')\varphi \, \mathrm{d}x = \int U'\varphi' \, \mathrm{d}x - [U'\varphi]$, and the bracket dies since $\varphi(0) = \varphi(1) = 0$.
→ $\langle U, \varphi \rangle = \int f\varphi$ ***is*** $\boldsymbol{K\xi = b}$**.**

$$K = \langle \varphi, \varphi \rangle = \int_0^1 (\pi \cos \pi x)^2 \, \mathrm{d}x = \frac{\pi^2}{2} = 4.934802, \qquad b = \int_0^1 2 \sin \pi x \, \mathrm{d}x = \frac{4}{\pi} = 1.273240,$$

$$\xi = b/K = 0.258012 = \frac{8}{\pi^3} \qquad \text{— the same number as above.}$$

→ u **obeys the same identity.** Computed two ways:

via u	$\displaystyle\int_0^1 (1 - 2x)\, \pi \cos \pi x \, \mathrm{d}x$	$= 1.273240$
via f	$\displaystyle\int_0^1 2 \sin \pi x \, \mathrm{d}x$	$= 1.273240$ ← **only f needed**

→ **Subtract.**

$$\begin{aligned} \langle u, \varphi \rangle &= 1.273240 \qquad u \text{ has no choice} \\ \langle U, \varphi \rangle &= 1.273240 \qquad = \boldsymbol{K\xi}, \text{ by construction} \\ \hline \langle e, \varphi \rangle &= \boxed{0} \end{aligned}$$

And yet the error is real:

x	$u(x)$	$U(x)$	$e(x)$
0.25	0.18750	0.18244	+0.00506
0.50	0.25000	0.25801	−0.00801
0.75	0.18750	0.18244	+0.00506

$$\langle e, e \rangle = \tfrac{1}{3} - \tfrac{32}{\pi^4} = 0.004822 \neq 0, \qquad \text{while} \qquad \langle e, \varphi \rangle = 0 \text{ exactly.}$$

Read it: you have *one* probe. It reads zero. The error is still there. With $M + 1$ basis functions it is the same sentence $M + 1$ times — and those $M + 1$ sentences are the rows of $\boldsymbol{K\xi = b}$.

13 The distinctions, side by side

term	what it is	zero?
$R(w)$	residual $-w'' + qw - f$, a function of x	$\neq 0$ for computable w
trial function	the candidate $U = \sum_j \xi_j \varphi_j$	—
test function	the probe v you integrate R against	—
N	shape function on one element	—
φ_i	global basis function at node i: the N's touching node i	—
Galerkin	the choice: test space = trial space	—
$\langle v, w \rangle$	$\int (v'w' + qvw)\,\mathrm{d}x$; appears after integrating $R\,v$ by parts	—
orthogonality	generic: $\langle a, b \rangle = 0$	—
K_{ij}	$\langle \varphi_j, \varphi_i \rangle$	$\neq 0$ — **that is the matrix**
Galerkin orth.	$\langle e, \varphi_i \rangle = 0$, $e = u - U$	$= 0$ — **that is the theorem**

14 The chain, in one line

R is the leftover $\longrightarrow$ cannot kill it everywhere (infinitely many conditions, finitely many unknowns) $\longrightarrow$ kill $M + 1$ *weighted averages* of it $\longrightarrow$ the weights are the **test functions** $\longrightarrow$ **Galerkin**: take them from the trial space, i.e. the φ's $\longrightarrow$ integrate by parts, and a **dot product** $\langle \cdot, \cdot \rangle$ appears $\longrightarrow$ $\langle U, \varphi_i \rangle = \int f \varphi_i$ *is* $\boldsymbol{K\xi = b}$ $\longrightarrow$ u obeys the same identity $\longrightarrow$ subtract $\longrightarrow$ $\langle e, \varphi_i \rangle = 0$: **Galerkin orthogonality**.